Sampling Methods KEY

Name _____

A school has 1,200 students. The principal wants to ask 60 of them about the new schedule. How should she choose the 60? The best methods use **chance** to pick, so no group is favored.

Method	How it works	Example (60 of 1,200)
Simple random sample (SRS)	Every group of the same size has the same chance of being chosen. Like drawing names from a hat.	Number the students 1 to 1,200. A random number generator picks 60 numbers.
Stratified	Split the population into groups (**strata**). People in the same group are alike in some way. Take a random sample from **each** group.	Randomly pick 15 students from each grade: 9th, 10th, 11th, 12th.
Cluster	Split the population into groups (**clusters**). Randomly pick some **whole** groups.	Randomly pick 3 homerooms. Ask everyone in them.
Systematic	Pick a random start. Then take every kth person. k is the population size divided by the sample size.	Start at a random student between 1 and 20. Then take every 20th student on the list.
Convenience (bad)	Take whoever is easy to reach.	Ask the first 60 students who walk in the door.

Stratified or cluster? Stratified: a **few** from **every** group. Cluster: **everyone** from a **few** groups.

Using a random digit table. Each digit is random. Read the digits in pairs for 2-digit numbers.

Row 1	52601 81590 83016 61318 60913 90996 03082 46281
Row 2	94821 99351 81909 37865 79754 32319 48757 49118
Row 3	62527 60189 55597 97114 71049 74650 75291 70342

Example: Choose 3 of 25 students, numbered 01 to 25. Use row 1.
Pairs: 52 60 18 15 90 83 01 . . .
Skip numbers over 25 and skip repeats. The sample is students **18, 15, and 01**.

Why random? Chance does not pick favorites. A random sample is not biased toward any group.

Work with a partner. Compare answers when you finish each problem.

1. Name the sampling method.

   (a) A store gives a survey to every 10th customer, starting with a randomly chosen customer from the first 10.
   Systematic

   (b) A city randomly picks 5 of its 40 neighborhoods and surveys every home in those 5.
   Cluster

   (c) A teacher puts all 28 names in a hat and draws 6.
   Simple random sample

   (d) A college randomly picks 50 first-year, 50 second-year, 50 third-year, and 50 fourth-year students.
   Stratified (the strata are the years)

   (e) A reporter asks people walking past a coffee shop.
   Convenience

2. A class has 30 students, numbered 01 to 30. Use **row 3** of the random digit table on page 1 to choose 4 students.

   Pairs: 62 52 76 **01** 89 55 59 79 71 **14** 71 **04** 97 46 50 75 **29**. The sample is 01, 14, 04, 29.

3. A school wants to survey 100 of its 1,000 students about sports. About half of the students play a sport.

   (a) Describe how to take a stratified sample.
   Split the students into athletes and non-athletes. Randomly choose 50 from each group.

   (b) Why might a stratified sample be better than an SRS here?
   It makes sure both groups are in the sample in the right numbers. An SRS could, by chance, pick too many athletes.

4. A farmer has 50 rows of corn. She wants to check 500 plants for insects. She randomly picks 5 rows and checks every plant in them.

   (a) What method is this? Cluster

   (b) Why would a farmer choose this method?
   It is quicker. She only walks 5 rows instead of searching the whole field.

5. Maya wants to know how students feel about school lunch. She surveys the first 40 students in the lunch line.

   (a) What method is this? What is the problem?
   Convenience. Students who skip lunch or come late are left out, so the results are biased.

   (b) Describe a better way for Maya to choose 40 students.
   Get a list of all students. Use a random number generator to pick 40 names (an SRS).